CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Results-oriented Software Engineer with 10+ years of experience designing and building high-performance applications, specializing in **C++ development for high-load and low-latency systems**. Skilled in optimizing algorithms, reducing execution bottlenecks, and developing **high-frequency trading (HFT) platforms** where every microsecond matters. Experienced in **cryptocurrency exchanges, networking protocols (REST, WebSockets, FIX)**, and distributed systems. Strong track record of delivering mission-critical trading features, ensuring stability under extreme traffic, and leveraging **AWS, GCP, Kafka, Kibana, Grafana, and Prometheus** for reliability and monitoring.

TECHNICAL SKILLS

- **Programming Languages:** C++ (11/14/17/20), Python, Java, C#, JavaScript, TypeScript, SQL, NoSQL
- **High-Performance Systems:** Multithreading, Concurrency, Lock-free data structures, Memory optimization, Low-latency tuning
- **Networking & Protocols:** REST, WebSockets, FIX, TCP/IP, UDP, ZeroMQ
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Cassandra
- **Cloud Platforms & DevOps:** AWS, GCP, Azure, Docker, Kubernetes, CI/CD, Terraform
- **Monitoring & Observability:** Kafka, Prometheus, Kibana, Grafana, ElasticSearch
- **Other:** Data structures, Design patterns, Algorithms, Distributed systems, Financial markets

WORK EXPERIENCE

Senior Software Engineer
Sensidev
September 2021 – December 2024
Timişoara, Romania (Remote)

- Engineered **C++ high-performance services** that handled thousands of parallel requests per second, optimizing market data processing pipelines for sub-millisecond response times under extreme load.
- Designed and maintained **low-latency trading algorithms** that connected to major cryptocurrency exchanges, reducing execution delays by 35% through optimized socket handling and custom data structures.
- Implemented **multithreaded order matching engine** with lock-free data queues, ensuring deterministic performance during high-frequency trading bursts and preventing latency spikes under 10µs.
- Collaborated with quantitative researchers to integrate **real-time price feeds** and ensure system accuracy, reliability, and resilience to exchange-side anomalies.

- Built **distributed data ingestion pipelines** across AWS/GCP using Kafka, enabling scalable processing of multi-gigabyte cryptocurrency tick data with zero downtime.
- Implemented **advanced monitoring and alerting dashboards** in Kibana and Grafana, improving incident detection speed by 45% and reducing mean time to resolution (MTTR).
- Enhanced **memory-management and cache locality** strategies in C++, lowering runtime memory footprint by 22% while maintaining ultra-fast execution throughput.
- Mentored engineers on **low-level performance profiling tools** (perf, gprof, Valgrind) and best practices in writing deterministic, high-frequency C++ applications.

Software Engineer
Syscom Digital
October 2019 – August 2021
Bucharest, Romania (Remote)

- Built **real-time analytics engines** in C++ and Python to process streaming financial and transactional data, delivering reliable insights with <5ms latency.
- Designed **networking modules with REST and WebSocket integrations** to ensure stable and fault-tolerant connections with third-party APIs under fluctuating traffic conditions.
- Contributed to **trading connectivity infrastructure**, ensuring exchange gateways were optimized for maximum throughput and resiliency against data loss.
- Optimized **thread synchronization primitives** to avoid bottlenecks and implemented fine-grained parallelism for computational workloads, boosting performance efficiency by 40%.
- Worked with **distributed systems architecture**, scaling applications to handle spikes of over 1M requests/hour across AWS clusters with automated failover.
- Integrated **Prometheus-based monitoring and Grafana dashboards** for latency visualization, proactively identifying issues before impacting system performance.
- Introduced **Kafka message queues** for inter-service communication, achieving fault tolerance and improving data streaming reliability during peak hours.
- Collaborated with DevOps teams to streamline **CI/CD pipelines** and ensured faster deployments with zero downtime across containerized services.

Software Developer
Soft Build
June 2016 – July 2019
Timişoara, Romania (Remote)

- Developed **multi-threaded C++ backends** for financial analytics and order management, enabling simultaneous handling of thousands of trade updates per second.
- Optimized **SQL/NoSQL hybrid database queries** for high-volume transaction systems, reducing data retrieval times by 28% under concurrent load.
- Implemented **socket-based communication layers** for real-time systems, ensuring consistent low-latency message delivery with guaranteed packet sequencing.
- Designed **data structure optimizations** that reduced memory fragmentation and improved application cache efficiency across high-load applications.
- Built **distributed computation frameworks** leveraging AWS Lambda and GCP Pub/Sub for large-scale data aggregation tasks.
- Applied **algorithmic optimizations** to ensure deterministic runtime execution, avoiding unpredictable stalls during peak throughput workloads.
- Integrated **trading simulation frameworks** for back-testing and validating execution strategies under high-frequency market conditions.
- Supported production environments by **debugging critical C++ runtime issues**, memory leaks, and race conditions affecting high-availability trading systems.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Created **modular backend services in C++ and PHP**, ensuring scalability for e-commerce and digital transaction systems.
- Designed **REST API gateways** optimized for concurrency and reliability, handling bursts of tens of thousands of client requests without downtime.
- Enhanced **multi-threaded job execution frameworks**, increasing parallel throughput by 33% while ensuring thread-safety and stability.
- Conducted **low-level profiling and latency benchmarking** to identify runtime hotspots and optimize server response times under load.
- Implemented **resilient error-handling and recovery strategies** for distributed systems with automatic retry logic for failed operations.
- Collaborated with cross-functional teams to deliver **performance-sensitive modules** that integrated with high-traffic third-party platforms.
- Improved **database query execution plans** using indexing and partitioning strategies, significantly reducing transaction times.
- Built **logging and monitoring solutions** for system observability, improving debugging accuracy and failure recovery speed.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Assisted in **C++ system development projects**, focusing on algorithmic optimizations and multi-threaded service reliability.
- Supported the design of **network protocol handlers** for REST and WebSocket-based messaging systems.
- Gained hands-on exposure to **low-level debugging and performance profiling**, resolving early bottlenecks in pre-production builds.
- Collaborated with senior developers to implement **core data structures and caching layers**, ensuring predictable performance under concurrent load.
- Contributed to **financial analytics prototypes** that simulated high-load trading behaviors using synthetic data sets.
- Implemented **basic monitoring dashboards** in Kibana to visualize system performance and identify anomaly patterns.
- Documented and tested **core modules for production release**, ensuring consistency and reliability across multiple deployment environments.
- Applied principles of **high-load systems engineering**, learning the foundations of HFT system design, networking, and concurrency.

EDUCATION

Tallinn University

Bachelor’s Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- GCP Associate Cloud Engineer
- C++ Institute – Advanced C++ Programming Certification
- Certified Kubernetes Application Developer (CKAD)
- Data Structures & Algorithms Specialization – Coursera

PROJECTS

- ✓ **Crypto Trading Engine:** Developed a **low-latency C++ trading engine** with sub-microsecond execution time, integrating FIX/REST/WebSocket protocols for multi-exchange connectivity.
- ✓ **High-Frequency Data Pipeline:** Built a **distributed Kafka-based ingestion system** for processing cryptocurrency tick data in real time with AWS/GCP scalability.
- ✓ **Latency Benchmark Suite:** Created **profiling and benchmarking tools** for optimizing concurrency primitives in trading systems, achieving consistent <10μs performance metrics.
- ✓ **Monitoring & Reliability Framework:** Implemented a **Grafana + Prometheus monitoring stack** to ensure observability, anomaly detection, and stability in HFT environments.